

A Simple, Secure and Energy Efficient Key Establishment Mechanism for Heterogeneous clusters of IoT devices in Smart Environments

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Abstract—With the extension of Internet and electronic commerce, the deployment of smart devices which have sensors and actuators in smart infrastructures is rapidly increasing. Most of the systems like heating, ventilation and air-conditioning systems, security systems, thermostat, sensors and other appliances have network connectivity and provide smart applications. These devices interact with other devices like gateways and work according to the commands given by the user devices. The user devices can be smartphones or computers (web interface). These communications should be secure enough to send and receive the commands, so that smart home or smart building appliances work appropriately. So, there is a need for a security mechanism to ensure that the communication between the devices is secure and there is no breach of authenticity, confidentiality and integrity. The goals of smart environments are to provide good quality of life, security, cost reduction, safeguard privacy, remote assistance, monitoring, energy management and convenience. In this paper, we are proposing a key establishment protocol for the clusters of devices in smart environments. The security analysis of the proposed protocol proves that it is secure and its performance analysis proves that it consumes 27% less energy as overhead than existing scheme.

Index Terms—Smart Environments, Internet of Things, Key Establishment, Security, Authentication, Heterogeneous clusters

I. INTRODUCTION

THE vast development in information society is playing a vital role in fueling the era of Internet of Things (IoT) over the last decade.

The rapid growth in telecommunication technologies is changing the way of living, working, interacting and even thinking. IoT / Machine-to-Machine (M2M) communication is the latest trend and it is leading the world towards the new opportunities and possibilities [5] [6]. Smart environments, like Smart Homes, Smart buildings, offices, hospitals, educational / industrial buildings and Smart cities are the focus areas for the development using M2M technologies [6].

In smart environment, heterogeneous network of IoT has IP based and non IP based smart devices which are connected through gateways (IP based). The gateway communicates with the server (or cloud based platform) and user devices via Internet. In this network, devices like sensors generate a huge amount of data which is transformed into operational data by the gateway using data analytics and integration for creating

intelligence. The gateway can also take certain decisions based on predefined local rules in case of some emergencies (critical conditions). It also monitors the health of the IoT devices and inform the call centres when these devices need to be replaced or repaired [22] [23].

Smart home network is an integral unit for bigger smart environment projects. Smart home market is classified into two categories, one is application based and other is technology based. From the perspective of application it is classified into Safety and Security, Energy Management Systems, Lightning Control Systems and Entertainment etc. On the basis of technology used, it is classified into Wired and Wireless that are further classified into Bluetooth, ZigBee, Z-Wave, Wi-SUN, Wi-Fi etc [1] [6] [10] [20].

As IoT/ M2M communication provides us the facility of monitoring and controlling Automotive, Power, Health, Safety and Security systems in smart environment, so there is transmission and processing of data / information and data analysis done by the devices to improve the efficiency of these systems. Devices in smart environment are small in size with limited battery and computation resources. Resource constraint, mobility and heterogeneity are major challenges of smart environments [3] [4] [5]. Confidentiality, authentication, access control, privacy, policy enforcement, trust, mobile security and secure middleware are security requirements of smart environments. Smart environments are more vulnerable to masquerade, man-in-the-middle, message forgery, replay and denial of service attacks [7] [14] [15] [19] [21]. The secure transmission of data and authentication of the devices with the gateway becomes predominantly necessary. Along with this, here we are concerned about the big organizations where there is need of more than one gateway as all IoT devices cannot be in the fair range of one gateway (most of the IoT devices have short range protocols) and there can be restrictions for accessing the IoT devices to different level of employees (users). As most of the IoT devices are resource constrained. So there is a need of secure and energy efficient mechanism which authenticates all the devices before adding them to the network and generates a session key between gateway and each IoT device to communicate securely.

This paper presents a simple, secure and energy efficient key

establishment mechanism for heterogeneous clusters of IoT devices in smart environments. The proposed work is proved to be secure and consumes 27% less energy than that of existing work [16].

II. LITERATURE REVIEW

In smart environments different kinds of IoT devices (sensors/actuators) are connected and most of the devices are smaller in size and have limited computational, power and communication resources. In 1975, when the smart home was designed by Pico electronics, USA, it was not done for connecting to the Internet. So the design of smart home was secure for in-home usage. As the utilization of automated systems increased, the smart homes were connected to Internet. Communication protocols for smart home IoT devices were not designed to be secure against intruders due to less memory and less computational capability, they cannot opt standard security protocols because they consume more processing power and memory. The pre-installed key management method does not work because there is risk of compromising the key in long term [13] [17]. So there arises a need of secure authentication and key establishment protocol which establishes a secure key and channel between the smart device and gateway, after the deployment of devices in the network.

In [1], the survey of different home automation protocol architectures from the perspective of their number of layers, security, end-to-end reliability, Internet connectivity and their implementation size is described. It also suggests IP-based (IPv6) solutions on which IETF is also focusing on the extension of Internet for sensors and actuators networks by standardizing the mechanisms. In [2], the focus is on health-care research discipline using telecommunications technologies and new automation industries. It further describes the concept of Data Mining for knowledge discovery and information transfer regarding health using TETRA networks. The projects [3], [4] and [5] (GENIO, SM4ALL and HOPE) are smart homes for elderly/disabled people which consists of numerous kinds of smart devices. The scheme proposed in [6] focuses on the security in terms of recognition of individual's regular activity pattern. It follows seamless distributed scheme and generates key-session using authentication token. Scheme [9] establishes the group key based on ECC (Eliptic-curve cryptography) [8] in Zigbee-based local HAN. A device authentication mechanism [9] was proposed for smart energy home area network. This mechanism is based on public key cryptography and consumes more energy and it is computationally complex. In [11], a device authentication scheme is proposed for HAN which uses ECC and self-certified public key technique [10]. The security analysis of the scheme is not given. In [12], key management protocol is proposed which takes the help of manufacturer for the authentication of consumer devices. The security analysis is done only theoretically, security is not verified by any simulation tool. In [16], session-key is generated securely using authentication token but a device can generate a session-key with a predefined gateway. In this scheme the device cannot be transferred to another gateway

if required. Another scheme is secure key distribution for the smart grid [18] which is based on public key cryptography. The communicating entities in this scheme are trusted third party (TTP), service provider (SP), and devices (smart meters). It also suffers from computation overhead and complexity. In [29] a group key is established between a group of nodes and gateway, the key is computed by nodes individually using group key parameters shared by gateway. The protocol provides confidentiality services to group messages and cannot be used for authentication of nodes. In [30] a group secret is established between a group of nodes initiated by a node (generally gateway), the group secret is shared among the group members and can be used to securely communicate between the group. It supports group communications and uses Elliptic Curve Diffie-Hellman key (ECDH) exchange mechanism to establish a group secret. The mechanism consumes more resources especially computational and cannot be used for authentication. Hence, we intend to design mechanism for secure key establishment, for large smart environments with multiple gateways.

III. PROPOSED SYSTEM

The proposed system design consists of system architecture and protocol for key establishment.

A. System Architecture

In this architecture (Fig.1), there are n number of group controllers (GC or M2M gateway) i.e. $GC_1, GC_2 \dots GC_n$ connected to the Server via broadband / FTTH (Fiber to the Home) or GSM 3G/ 4G. Here, server is a dedicated entity that connects a smart environment to outside world and also performs complex computing operations on the data. The server can be as usual a secure device. The smart devices in Home Area Network or Local Area Network are connected with GC by through low power wireless communication technologies- Bluetooth Low Energy, Z-wave, 6LoWPAN, ZigBee, Wi-Fi etc. Group Controller (GC) acts as a mediator between server and smart devices (SD). It also passes the commands from the users to the SDs and response from the SDs to the user devices. There can be any number of smart devices ($SD_1, SD_2 \dots SD_n$) connected to every GC. GC is a intermediate device like an Internet Access Point in a Wi-Fi network. It is generally not compromised. An attacker tries to attack a communication between GC and SD. GC may be connected to server via wired network. The goal is to establish a secure key between each SD and respective GC, so that the SDs can be operated only through commands from their respective GCs. The proposed architecture supports heterogeneous clusters, whereas the previous works supports only a single cluster of IoT devices, but the proposed architecture includes many clusters through GC's. Each cluster may be different from others in terms of architecture, functionalities, deployment environment etc. These differences are resolved through GC's and enables migration of SD's between clusters. When a new device is added to any group, then it should be able to establish a key with the respective GC (scalability). If any device is shifted from one group to another group, then it should be

able to generate a new key with the new GC and its IP address changes accordingly. The establishment is heterogeneous as it enables different types of clusters of SD's to be connected through their GC's and also supports mobility of SD's between clusters.

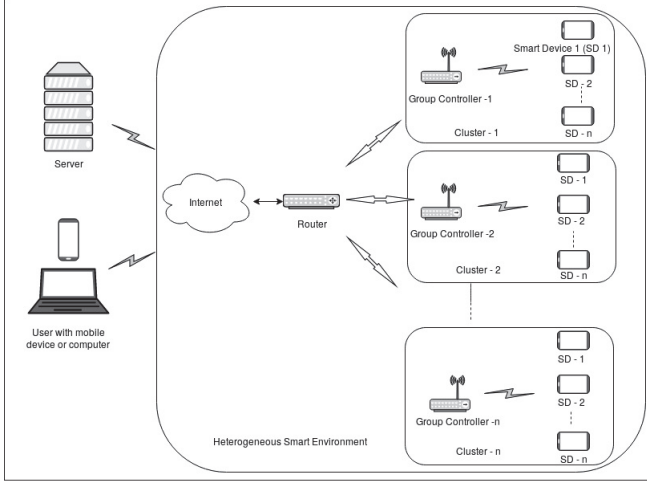


Fig. 1. Proposed System Architecture

B. Key Establishment Protocol

In this scheme there are two phases for key establishment. In the first phase, some initial values (generator a with cyclic group p , secret key K) are stored in the storage of GCs by the service provider or server. In the second phase, authentication and key establishment is done between the SD and respective GC. The table I shows the symbols and description that are used in the protocol. It is assumed that the Server (S) and the GC are reliable entities and are connected through secure channel [16]. The GCs and the SDs have same encryption, decryption, hash and MAC functions. Clocks of all the GCs and SDs are synchronized using the scheme of Li et al. [26], and are agreed on a transmission delay (ΔT) to avoid replay attacks.

C. Setup Phase

TABLE I
SYMBOLS AND DESCRIPTIONS

Symbol	Description
S	Server
Id_A	Identity of Smart Device A
Id_{G_i}	Identity of Group Controller i
M_K	Message M is encrypted with key K
$Token_A$	Authentication token for device A
$MAC()$	Message authentication code
$\parallel$	Concatenation operation
a	generator
T_i	Timestamp i
r_i	Random number of group controller i
Q	Secret value sent by Server to device

- Prior to the deployment the server assigns identity of every GC (Id_{G_i}) and stores a random number r_i , a

generator a with cyclic group p and a secret key (K_{G_i}) between the server and GC in the memory of the GC through a secure channel.

Server (S) $\rightarrow GC_i : Id_{G_i}, K_{G_i}, r_i, a, p$

- For each SD say device A the server assigns identity of A (Id_A) and stores all the group controller's IDs and calculates R_i (hash of the identity of device A, identity and random number of GC), a short authentication token ($Token_A$), a generator a over a cyclic group p (same as group controllers) and a secret key with the server (K_A) in the memory of the device.

S $\rightarrow A : Id_A, K_A, Id_{G_1}, Id_{G_2}, \dots, Id_{G_n}, R_1, R_2, \dots, R_n, Token_A, a, p.$

where $R_i = h(Id_A \parallel Id_{G_i} \parallel r_i)$

The server will maintain a database of all the deployed devices. Fig. 2 shows the block diagram of the proposed protocol.

D. Authentication and Key Establishment phase

- Smart device A wants to establish a key with $GroupController_i (GC_i)$; which is new GC of device A, so it sends message 1 to GC_i .

A selects a random number x and computes a^x . T1 is the current timestamp of Device A. It encrypts group controller's id, its own id, $Token_A$ and a^x by using the secret key with sever (K_A). We name this component as P. It also computes MAC of Id_A , T1, P and Id_{G_i} using key R_i , so that the integrity and authenticity of the message can be ensured by GC_i . Here all the exponentials are in a cyclic group p .

Message 1 : A $\rightarrow GC_i : Id_A, T1, P, MAC[R_i, Id_A \parallel T1 \parallel P \parallel Id_{G_i}]$

Where $P = \{Id_A, Id_{G_i}, Token_A, a^x\}_{K_A}$

- When the request comes to GC_i , it first compares the delay with mutually agreed transmission delay. If the delay is less than or equal to $\Delta T = (T2 - T1)$ ($T2$ is the current time stamp of GC_i), then it will check the integrity of the message otherwise it will reject the request. It computes the key R_i by calculating hash of the random number r_i , identities of itself and requesting device A, then it calculates MAC of the message and compares it with the received MAC. If the received and calculated MAC is equal then only it will further process the request. Thus, it will be sure that the request by the device A is for GC_i only. Then, GC_i sends message 2 to the server.

Message 2 : $GC_i \rightarrow S : Id_{G_i}, \{P, Id_A, a^y, T2\}_{K_{G_i}}$

In this request GC_i encrypts P, Id_A and a^y that are received from device A and current timestamp T2 by using secret key with Server.

- When the server receives the message, it decrypts with K_{G_i} . First it compares the delay between timestamp T2 and current timestamp T3. After ensuring that the message is current, it will process further. It decrypts the component P and gets $Id_A, Id_{G_i}, Token_A$ and a^x . It compares $Id_A, Token_A$ and Id_{G_i} with the stored data. If everything is correct, then the server will be sure that

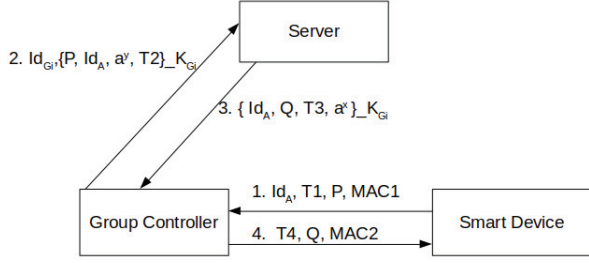


Fig. 2. Block diagram of proposed protocol

the request for the key establishment has come to GC_i from device A only. Server authenticates the device A and sends reply message 3 to GC_i .

Message 3 : $S \rightarrow GC_i : \{Id_A, Q, a^x, T3\}_{K_{Gi}}$

Where $Q = \{Id_A, Id_{Gi}, Token_A, a^x, a^y\}_{K_A}$

The reply contains id of device A, current time stamp and Q (i.e. $Id_A, Id_{Gi}, Token_A, a^x, a^y$ encrypted with K_A) encrypted with K_{Gi} .

- The reply from the server is decrypted by GC_i with K_{Gi} . GC_i compares transmission delay with ΔT and Id_A , then it will be sure that device A is legitimate entity. It accepts a^x and generates a random number y and computes key $K = (a^x)^y$. It also computes a^y and sends Q along with the response.

Message 4 : $GC_i \rightarrow A : T4, Q, MAC[Id_A || Id_{Gi} || T4 || Q || T1]$

The response contains MAC of $Id_A, Id_{Gi}, T4, Q$ and timestamp of the request message from the device A (i.e. $T1$) with key R_i .

- When device A receives the response from GC_i , first it will check the timestamp. Then, it will calculate MAC to check the integrity of the message and then decrypts the component Q with secret key K_A and gets its own id, $Token_A$, GC_i 's id, a^x and a^y which it sent before for key generation. If all the details are correct, it will know that a^x is accepted by the GC_i and now it will also calculate the key $K = (a^y)^x$. Thus the key K is established between SD say device A and Group controller i say GC_i .

As shown in the flow chart of the proposed protocol in fig 3. The complete process of authentication and key establishment is done in four passes. The device A is required to communicate in only two passes. Thus, this protocol minimizes the energy consumed by device A. The proposed protocol is broadly immune to following security attacks:

- Replay attack: The messages exchanged during authentication and key establishment contains timestamps, which avoids replay attacks and subsequently Denial of Service (DoS) attacks.
- Masquerade attack: The exchange of id's in encrypted form, timestamps and MAC avoids masquerade attack.
- Sybil attack: In every message exchanged, id of a device is encrypted and moreover a sender computes and sends MAC and receiver verifies MAC during messages 1 and 4, this avoids Sybil attack.

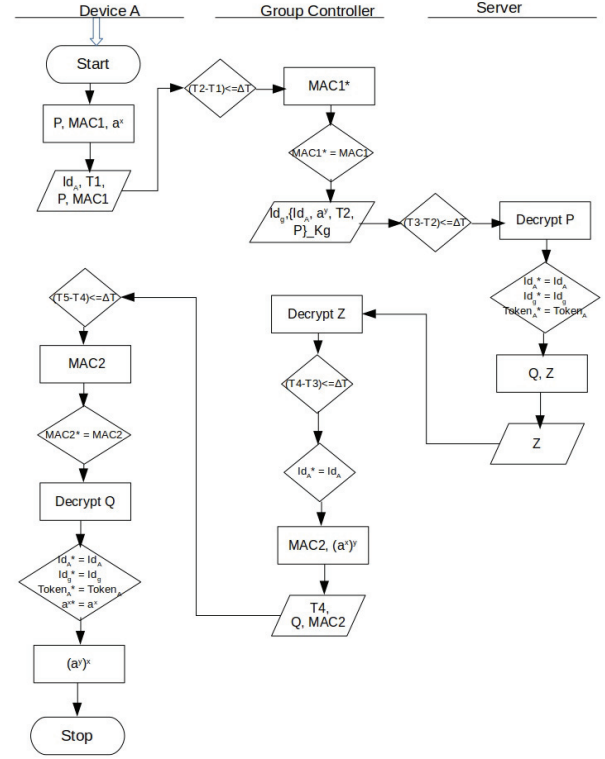


Fig. 3. Flow chart of proposed protocol

- Man-in-the-middle attack: Messages are encrypted using secret keys between communication devices thereby avoiding man-in-the-middle attack.
- Message Delay attack: The exchange and verification of timestamps in messages avoid delay attacks.

IV. AUTOMATED SECURITY ANALYSIS AND VERIFICATION USING SPAN+AVISPA TOOL

For the security analysis of the protocol, modeling is done with SPAN + Avispa-1.6 tool. It verifies the security of the protocol against Dolev-Yao channel. In this channel all the messages are seen to the intruder which can modify the messages. SPAN is Security Protocol ANimator for AVISPA which animates all the transitions between the entities. There can be any number of entities in protocol model. HLPSL language (High Level Protocol Specification Language) is used for protocol modeling. Each entity's role along with its transitions, is defined separately.

In this protocol model three roles are defined which are deviceA, devGty and devServPro. The three roles deviceA, devGty and devServPro are played by agents A, DG1 and DSP respectively. Agent A knows K_a as its shared secret key with server. Agent DG1 knows K_{dg} as its shared secret key with server. Agent DSP is with keys K_a and K_{dg} . Device A and Group Controller (devGty) have same generating element

'a'. The initiator is deviceA which is initialised with the state 0. devGty and devServPro start with state 1 and 2 respectively. In role environment, Intruder knowledge and session compositions are defined. The goal section has definition of goals for authentication and secrecy of entity or data.

A. Modeling of proposed protocol

The proposed key establishment protocol is modeled in HLPSL, the coding consists of five modules where first three modules are of device A, GC and server, forth module is of sessions & environment and fifth module defines goals of the protocol. First, second and third modules contains HLPSL code for device A, GC and server respectively which contains the definition of data including keys, its initial state and transitions (changes in its states through communications with other entities). In module 4 the sessions between the entities, intruder knowledge, send and receive operations are defined. In module 5 desired goals of the proposed protocol are defined. Two types of goals are defined, first is secrecy of values ($Token_A$, x and y) and second is authentication of devices. The HLPSL code is run in Avispa with OFMC backend. In case the protocol runs completely it means there are no errors in the protocol and all the transitions are completed.

B. Results generated

The protocol modeled in HLPSL, is simulated in SPAN+Avispa with back-end OFMC. The HLPSL code is taken as input by the HLPSL2IF converter which converts the HLPSL language model into intermediate format. Then this code works as input for the OFMC back-end. It runs the model for bounded number of times in sessions and checks if any attack is possible or not by the intruder. The HLPSL code is run in Avispa with back-end OFMC and the simulation is viewed using SPAN+AVISPA.

Module 1 : Pseudo code for role A (Smart device A)

```
Inputs
DSP, DG1, A: agent,
Kd : symmetric_key,
IdA, IdDG1 : text,
TokenA : text,
G: nat,
HidA : text,
HASHA : hash_func,
SND_DSPA, RCV_DSPA, SND_DGA, RCV_DGA:
channel (dy)
Read
State : nat,
T1,T4: text,
X : nat,
S : message
State = 0 /\ RCV_DGA(start)
SND_DGA (IdA.T1'.{IdA.IdDG1.TokenA.
exp(G,X')}_Ka.HASHA(IdA.IdDG1.T1'.
{IdA.IdDG1.TokenA. exp(G,X')}_Ka))
```

```
secrecy of TokenA
secrecy of X
witness of IdDG1
witness of TokenA
State = 3 /\ RCV(T4'.
{IdA.IdDG1.TokenA.S'.exp(G,X)}_Ka
.HASHA({IdA.IdDG1.TokenA.S'.exp(G,X)}_Ka.
T4'.IdA.IdDG1.T1)) = | >
end role
```

Module 2 : Pseudo code for role device gateway (Group Controller)

```
Inputs
DSP, DG1, A: agent,
Kdg : symmetric_key,
IdDG1 : text,
G: nat,
HASHA : hash_func,
SND_DSPDG, RCV_DSPDG, SND_ADG,
RCV_ADG: channel (dy)
Read
State : nat,
T1, T2, T3, T4: text,
Y : nat,
Q : message,
TokenA, IdA: text,
P,R : message
State = 1 /\
RCV_ADG(IdA'.T1'.P'.
HASHA(IdA'.IdDG1.T1'.P')) =| >
SND_DSPDG(DG1.{IdA'.exp(G,Y').P'.T2'}_Kdg)
request of IdDG1
witness of IdA)
secrecy of IdDG1
secrecy of Y)
State = 4 /\
RCV_DSPDG(DSP.{IdA.Q'.T3'.R'}_Kdg) =| >
SND_ADG(T4'.
Q'.HASHA(Q'.T4'.IdA.IdDG1.T1))
end role
```

Module 3 : Pseudo code for role devServPro (Server)

```
Inputs
DSP, DG1, A: agent,
Kdg, Ka: symmetric_key,
IdDG1, IdA: text,
TokenA: text,
HidA : text,
HASHA : hash_func,
SND_DGDSP, RCV_DGDSP, SND_ADSP,
RCV_ADSP: channel (dy)
Read
State : nat,
T2, T3: text,
U,V : message
```

State = 2 /\RCV_DGDSP(DG1.{IdA.U'.
 {IdA.IdDG1.TokenA.V'}_Ka.T2'}_Kdg)
 SND_DGDSP(DSP.{IdA.{IdA.IdDG1.TokenA.
 U'.V'}_Ka.T3'.V'}_Kdg)
 request of IdA
 request of TokenA
 end role

Module 4 : Pseudo code for role session and environment

Inputs

DSP, DG1, A: agent,
 Ka, Kdg: symmetric_key,
 G: nat,
 HASHA: hash_func,
 IdA,HidA,IdDG1,TokenA : text
 Read

SND_DSPA, RCV_DSPA, SND_DGA, RCV_DGA,
 SND_DGDSP, RCV_DGDSP, SND_ADSP, RCV_ADSP,
 SND_DSPDG, RCV_DSPDG, SND_ADG, RCV_ADG:
 channel (dy)
 composition =
 deviceA(DSP, DG1, A,Ka,IdA,
 IdDG1,TokenA, G,HidA, HASHA, SND_DSPA,
 RCV_DSPA, SND_DGA, RCV_DGA)
 /\devGty(DSP, DG1, A,Kdg,IdDG1,G,HASHA,
 SND_DSPDG, RCV_DSPDG, SND_ADG, RCV_ADG)
 /\devServPro(DSP, DG1, A,Kdg, Ka,IdDG1,
 IdA,TokenA,HidA,HASHA,SND_DGDSP,
 RCV_DGDSP, SND_ADSP, RCV_ADSP)
 end role

Module 5 : Pseudo code for goals declaration

Goals

secrecy_of TokenA t1
 secrecy of value X t2
 secrecy of value Y t3
 secrecy of IdDG1 t4
 authentication on IdDG1 t12
 authentication on IdA t13
 authentication on TokenA t14

end goal

Fig.4 shows the outcome of the protocol run in security analysis tool Avispa. The protocol run was within the bounded number of sessions defined in module 3, the protocol run could reach the goals specified in module 5 of HLPSSL code. The summary of the protocol run by Avispa is "SAFE", it means Avispa could not detect any security attack during protocol run and hence it is safe. The intruder had knowledge of all possible public values, the intruder is Dolev-Yao model. The protocol runs safely and fulfills all goals which were specified in the goal section. The simulation results shows that the protocol runs completely and is safe i.e. secure.

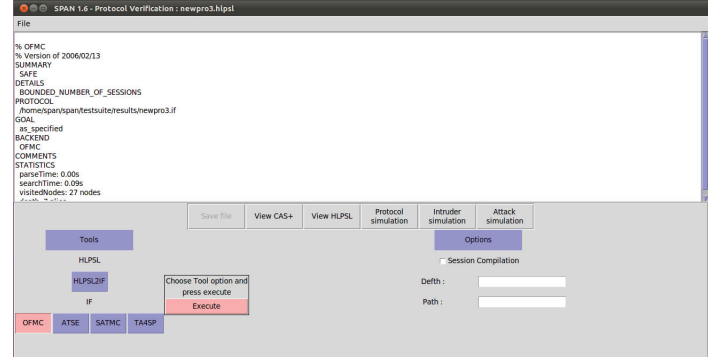


Fig. 4. Protocol Verification

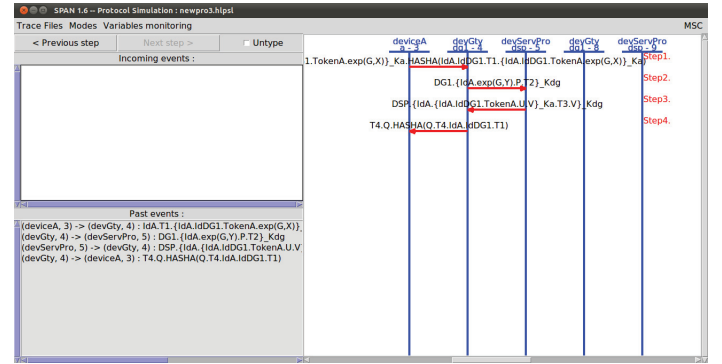


Fig. 5. Protocol Simulation

Fig.5 Shows the protocol simulation. It is visible that all the transitions between the entities are in order and complete. It means that the protocol run is successful and complete. Therefore there are no errors in the protocol. It may be concluded that the protocol is complete, error free and secure.

V. PERFORMANCE ANALYSIS

The GC and the server have sufficient power and computational resources. Whereas SDs have limited power, storage and computational resources due to their size and components cost. Any mechanism designed for smart environments should consume less energy specially with respect to SD. Hence to evaluate the performance we calculate security overhead at SD by the proposed protocol. To compute the security overhead we calculated computation cost overhead and communication cost overhead. In computation there are four types of operations, modular exponent, MAC, encryption and decryption. The size is taken as follows : ID 1 byte, timestamp 4 bytes, token size 2 bytes, modular exponent 2 bytes, AES 16 bytes and MAC 16 bytes. The send request (1st message) from node A has 296 bits [(1+4+16+16)bytes] and the receive message (4th message) has 288 bits [(16+4+16)bytes] . According to [16] the energy consumed for transmitting 1 bit is 0.72 μ J and for receiving 1 bit is 0.81 μ J. So the energy consumption by the node A is 446.40 μ J (transmitting 213.12 μ J and receiving 233.28 μ J). This is communication overhead of node A. The computation overhead is calculated by $E=V*I*T$ where V is voltage, I is current and T is time. For teleosb V is 3 volt and I

is 1.8 mA. The time taken for AES encryption is 3.6 ms , CBC-MAC 8.4 ms [16][25] and for modular exponent calculation it is 0.074 ms. The total energy overhead for computation is 65.139 μ J which is very less than the schemes described in literature. Table II shows the energy cost of performing cryptographic operations at device A and Table III shows communication costs of send and receive operations at device A. Fig. 6 shows the communication overhead at a device for various schemes, it is visible that the proposed scheme incurs less communication overhead than others. Fig. 7 shows the time consumed at a node for key establishment under various schemes. It is visible that the proposed scheme takes least time for key establishment. Fig. 8 shows the energy consumed at a node for key establishment under various schemes. It is visible that the proposed scheme consumes least energy among all.

TABLE II
ENERGY COSTS FOR CRYPTOGRAPHIC OPERATIONS

Operation at device A	Energy Cost (in micro-Joule)
Encryption	19.44
CBC-MAC	45.36
Modular Exponent	0.399
Total Energy	65.199

TABLE III
COMMUNICATION ENERGY COST

Messages at device A (in bits)	Energy Cost (in micro-Joule)
Send request (296)	213.12
Receive (288)	233.28
Total Energy	446.40

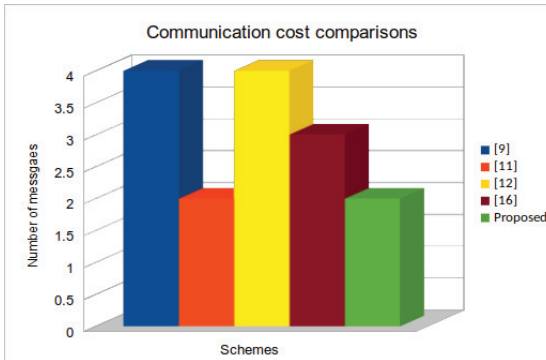


Fig. 6. Communication cost comparison in terms of number of messages exchanged with smart device

TABLE IV
COMPUTATION COST COMPARISON

	[9]	[11]	[12]	[16]	Proposed
Point Multiplication	2t	3t	-	-	-
Modular Exponent	-	-	-	-	2T
Hash Operation	1H	4H	5H	2H	-
MAC	1 MAC	-	7 MAC	1 MAC	2 MAC
HMAC	-	-	-	1 HMAC	-
Cryptosystem	1E+1D	-	4E+4D	1E+1D	1E+1D
Total cost (in ms)	1013.36	1591.04	376.4	130.11	94.67

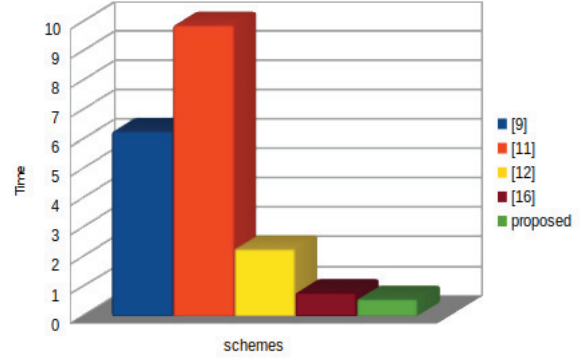


Fig. 7. Time comparison

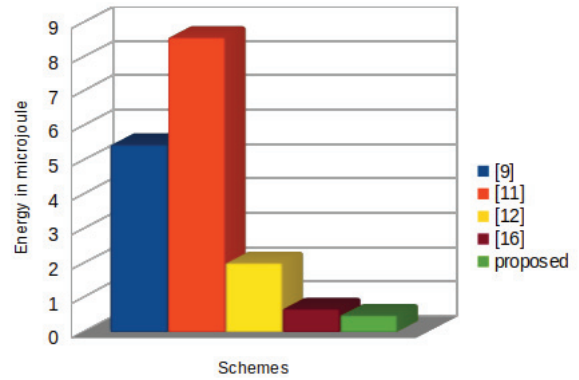


Fig. 8. Energy comparison

In table IV, t is the time to calculate point multiplication (480 ms) [27], T is the time to calculate modular exponent (0.074 ms), H is the time for hash operation (37.76 ms) [16], MAC is the time to calculate MAC (8.4 ms), $HMAC$ is the time to calculate hashed MAC (39 ms), E is the time for AES encryption (3.6 ms) and D is the time for decryption (3.6 ms) [26] [28]. In the proposed scheme, as there are total 4 passes and there are only two passes with the device A. So, the pass complexity of device A is two. The computation by device A takes 2 power operations, 2 MAC, 1 encryption and 1 decryption time, which is less than the previous proposed schemes. The total security overhead of the work in [16] is 702.62 μ J whereas the security overhead of proposed scheme is 511.22 μ J. The proposed mechanism not only provides key-establishment scheme for heterogeneous smart environments through multiple clusters but also achieves the desired outcomes at the cost of 27% less energy than most recent existing scheme [16]. The mechanism uses simple operations like exponential, group operations, MAC and symmetric key encryption moreover the number of messages exchanged are less thereby it consumes less resources and burdens with less overhead without compromising on performance and security. This scheme is light-weighted from the perspective of communication and computation with device A.

VI. CONCLUSION

The security in automation systems is the biggest challenge in today's world, as they are used in day to day life for safety, security, convenience and energy management. Smart environments have eased the life of individuals and their usage is growing enormously with time. These systems use many different kinds of small IoT devices which are resource constrained and they are more vulnerable to attacks. So, in this paper we proposed an energy efficient and secure key establishment mechanism which authenticates the smart devices before connecting to the GC/gateway and then establishes a key between the GC/gateway and smart device. In the proposed work, a key is established between smart devices and gateway. the key is computed by a smart device on its own and it uses the value from gateway. A smart device is authenticated before key establishment to avoid unauthorized access of the resources. The mechanism also ensures secrecy, avoiding many possible attacks. Moreover, it consumes around 27% less overhead to achieve its objectives. This mechanism is light-weighted, efficient and consumes less energy than the proposed schemes in literature.

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